

SUMMARY

AI/ML Researcher and Generative AI Application Developer with hands-on experience spanning Federated Learning, Computer Vision, Deepfake Detection, LLM Security, and Reinforcement Learning. Research work currently under peer review at Springer, with an Indian patent filed for an IoT-enabled computer vision system. On the engineering side, experienced in fine-tuning and deploying open-source LLMs into production using Python, FastAPI, and MLOps tooling (Docker, Kubernetes, CI/CD). Applies a research-grounded approach to building reliable, scalable AI systems – from experimental pipelines to production deployments. Author of peer-reviewed research (Springer), Indian Patent holder, and 2x Smart India Hackathon National Finalist.

EXPERIENCE

- 8/2025 – 5/2026

Associate Software Engineer – Generative AI & Applied Research

OpenXcell Technolabs, Ahmedabad

 - Researched and implemented an LLM Security Suite (Red/Blue Teaming) using LangGraph agent orchestration with NVIDIA Garak, LLM Guard, and Guardrails AI; evaluated prompt injection and jailbreak attack vectors systematically, mitigating 90%+ vulnerabilities across production use cases with AI ethics and bias mitigation considerations.
 - Designed and built RAG-based NLP pipelines with vector embeddings for text generation and summarization; researched retrieval strategies and embedding models to reduce hallucinations, and developed conversational AI chatbots integrated with business applications.
 - Deployed AI models via MLOps pipelines (Docker, Kubernetes, CI/CD) with SRE monitoring tools (Prometheus, Grafana) and incident response alerting to maintain reliability and high availability of production AI services.
- 6/2025 – 7/2025

AI/ML Research Intern – Medical Signal Processing

IEEE EMBS Pune Chapter, Pune

 - Conducted research on deep learning approaches for EEG-based Schizophrenia detection; designed and evaluated ResNet50, EfficientNetB2, and DenseNet121 architectures for accuracy-latency trade-offs using PyTorch, TensorFlow, and Scikit-learn.
 - Achieved 96% classification accuracy via ensemble stacking, improving model robustness and generalizability for clinical deployment – outcome documented as a research deliverable for the IEEE EMBS chapter.
- 8/2024 – 11/2024

Software Development Intern (Full-Stack & Data Engineering)

Institute for Plasma Research, Gandhinagar

 - Migrated legacy systems to a full-stack web platform using Node.js and TypeScript, and built real-time data engineering and analytics pipelines (Pandas, NumPy) transforming raw experimental data into actionable monitoring dashboards.

RESEARCH & PUBLICATIONS

- Springer Nature

FED-DETR: Privacy-Preserving Intelligent Traffic Enforcement

Peer Review

Developed a decentralized object detection pipeline using Federated Learning and DETR (Detection Transformer) deep learning for automated helmet detection in intelligent traffic monitoring. Currently under peer review at the International Journal of Intelligent Transportation Systems Research (Springer).
- Indian Patent

Automated Waste Segregation Dustbin for Educational Institutions

Under Review

Designed and prototyped an IoT-enabled smart dustbin integrating a lightweight computer vision model for real-time automated waste classification, optimizing ML model inference for edge computing. Application No. 202521111515, verifiable at iprsearch.ipindia.gov.in.

EDUCATION

- 2022 – 2026

LDRP Institute of Technology and Research, Gandhinagar

B.E. Information Technology – CGPA: 7.74/10
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Software Engineering.

PROJECTS

- Finalist

Incident Response Commander – AI Agent RL Environment (SRE)

Meta x Scaler OpenEnv 2026 – Project Link

 - Designed a novel Reinforcement Learning research environment simulating SRE incident response across a 10-service microservice architecture; formulated a multi-dimensional reward function covering investigation, root cause identification, remediation, time efficiency, and safety – submitted as a reproducible benchmark for LLM agent evaluation.
 - Implemented a Python/FastAPI backend with a structured observation-action loop, Dockerized on HuggingFace Spaces, and benchmarked a Llama 3.1 8B baseline agent scoring 0.70/0.62/0.56 across Easy/Medium/Hard tasks; selected in the top 800 of 31,000+ participants nationwide.
- Finalist

Face-Swap Deepfake Detection – Research Platform

Smart India Hackathon 2024 – Project Link

 - Researched ensemble deep learning approaches for face-swap deepfake detection, evaluating EfficientNet, InceptionNetV3, and Vision Transformers (ViT) across 5 benchmark datasets (FaceForensics++, Celeb-DF, DFDC, DFD, WLDR) totaling 82.6 GB, plus a custom-built dataset for real-world testing.
 - Deployed the detection system as a FastAPI microservice with Celery and RabbitMQ for async video analysis, bridging the gap between research experimentation and production-ready AI deployment.

Runner-Up	Qubits Learnova – eLearning Platform Odoo × GVP National Hackathon 2026 – Project Link
	- Architected a modular Express.js backend with Prisma ORM and PostgreSQL covering 13 lesson types, multi-role RBAC, Razorpay payment integration with HMAC signature verification, gamification, and AI quiz generation using LLMs. - Built a standalone video microservice using FFmpeg and BullMQ for async HLS transcoding into 360p/720p/1080p adaptive streams with S3/MinIO storage, webhook callbacks, and CI/CD via GitHub Actions and Docker Compose.

GLOBAL EXPOSURE & INTERNATIONAL PROGRAMS

Oct 2025	UK–France Immersion Future Leaders Program 2025 University of Kent (UK) & IESEG School of Management (France)
	- Selected for a competitive 1-month hybrid international immersion program under SVKM Trust & Kadi Sarva Vishwavidyalaya (KSV), focusing on global leadership and cross-cultural collaboration. - Completed 6-day academic immersion at the University of Kent (London) and a 6-day business residency at IESEG School of Management (Paris), including a Custom Course in Luxury Marketing & International Business Strategy, supplemented by a 15-day structured online global component.

ACHIEVEMENTS & PRESENTATIONS

2026	Finalist – Meta × Scaler OpenEnv Hackathon 2026 Ranked among the top 800 finalist teams out of 31,000+ participants nationwide, advancing to the grand finale by tackling advanced Reinforcement Learning problems.
2026	Runner-Up – Odoo × GVP National Hackathon 2026 Secured 2nd position in the national-level competition, competing against top engineering teams across India.
2026	Qualified – GATE DA 2026 Successfully cleared the Graduate Aptitude Test in Engineering for Data Science & Artificial Intelligence.
2025	2nd Runner-Up – SSIP State Level Hackathon Ranked 3rd among numerous teams for developing impactful technical solutions under the SSIP initiative.
2025	2x Finalist – Odoo National Hackathon 2025 (GVP & IITGN) Recognized as a national finalist in both the March and June editions of the Odoo National Hackathon.
2025	GTU National Poster Competition – AI-Powered Research & Innovation LinkedIn Post Presented a research poster at GTU's national-level competition on AI-powered innovation, communicating ML research outcomes to an academic and industry audience.
2024	VSITR State-Level Poster Competition – Autonomous Vehicles & AI LinkedIn Post Presented AI-driven autonomous vehicle research at the VSITR state-level poster competition, covering perception pipeline and edge ML deployment approaches.
2023 & 2024	2x Finalist – National Level Smart India Hackathon (SIH) Selected as a national finalist for two consecutive years, competing among the top innovation teams in India.

LEADERSHIP

2025	Student Organizer – ImpactThon & ERTE 2025 Led end-to-end execution of GUJCOST-sponsored University and State-level technical events, coordinating speakers, participants, and logistics.
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2024	Webmaster – IEEE LDRP-ITR Student Branch Managed and maintained the official web presence and digital infrastructure for the IEEE LDRP-ITR SB community throughout 2024.

SKILLS

- Generative AI & NLP:** LLM Fine-Tuning & Optimization, RAG Pipeline Development, Conversational AI & Chatbot Development, Natural Language Processing (NLP), Text Generation, Summarization, Sentiment Analysis; Frameworks: LangChain, LangGraph, AutoGen, Hugging Face, OpenAI GPT, spaCy, NLTK.
- ML / DL & Research:** Deep Learning, Computer Vision, Federated Learning, Reinforcement Learning, Ensemble Methods, Model Evaluation & Benchmarking; Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, SciPy.
- AI Deployment & MLOps:** FastAPI, Docker, Kubernetes, CI/CD (GitHub Actions), Production Environments, High Availability, Software Design Principles, AI Ethics & Bias Mitigation.
- SRE & Monitoring** (Familiar with): Prometheus, Grafana, ELK Stack, Incident Response Automation, Reliability Engineering, Alerting Pipeline Development.
- Cloud & Infrastructure:** AWS (Cloud Practitioner), GCP, Azure, Terraform, Ansible, PostgreSQL, MongoDB, Redis, Nginx.
- Programming & Tools:** Python, C++, TypeScript, JavaScript, SQL, Node.js, Git, GitHub, Figma.
- Soft Skills:** Research Communication, Proactive Problem-Solving, Analysis, Cross-functional Collaboration, Leadership.

CERTIFICATIONS

CLF-C02	AWS Certified Cloud Practitioner	Amazon Web Services
NPTEL	Database Management System	NPTEL